Waste collection VRP with stochastic time dependent demand increment

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Abstract

In this work..

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1. Introduction

We address the Waste Collection Vehicle Routing Problem with Stochastic Time-Dependent Demand Increment. In our problem a set of collection points must be visited by a set of identical capacitated vehicles located at a common depot. Each collection point has a known initial quantity of waste. This quantity may increase along the day. We divide the planning horizon

(day) in homogeneous not-overlapping time-slots or periods. For every collection point and every time-slot, the demand increment is uncertain. We represent this increment as a random variable. We assume that, if the residual capacity of the vehicle is sufficient, the vehicle loads all the waste, otherwise, it partially loads the waste or totally skips the node. The goal is to minimize the expected total cost given by net routing costs and penalties associated with uncollected waste at the end of the planning horizon. Note that high quantities of waste left over for the next day represent an undesirable event since they cause inconvenience to the citizens. For this problem, we provide a two-stage stochastic formulation.

2. Mathematical Formulation

2.1. Stochastic Model with Skipping

Let m be the number of available vehicles and G a direct graph. Nodes $1, ..., n$ of G represent the collection points, whereas node 0 and nodes $n + 1, ..., n + m$ represent the central depot (same location). In practice, there exist copies of a same depot; 0 represents the starting point of all routes and $n + h$ $(h = 1, ..., m)$ represents the final node of route h. In G there are only outgoing arcs from node 0 towards all other nodes and only entering arcs in nodes $n + 1, ..., n + m$ from all other nodes. A collection point $j \in \{1, ..., n\}$ has outgoing arcs towards and entering arcs from all other nodes. In order to present the formulation, we introduce the following additional notation for sets, input, indices, and decision variables.

Sets:

- $V = \{0, 1, ..., n, n + 1, ...n + m\}$: set of all nodes
- $V' = \{0, 1, ..., n\} = V \setminus \{n + 1, ..., n + m\}$
- $V'' = \{1, ..., n, n + 1, ...n + m\} = V \setminus \{0\}$
- S : set of scenarios
- T : set of periods

Indices:

- i, j : indices of node

- t : index of period
- s : index of scenario

Input:

- C : vehicle capacity
- c_{ij} : travel cost associated with arc (i, j), $i \in V'$, $j \in V''$ $(j \neq i)$. Since depot 0 physically coincides with nodes $n+1, ..., n+m$, then $c_{0j} = 0$ for $j = n+1, ..., n+m$
- t_{ij} : travel time associated with arc (i, j), $i \in V'$, $j \in V''$ $(j \neq i)$. Since depot 0 physically coincides with nodes $n+1, ..., n+m$, then $t_{0j} = 0$ for $j = n+1, ..., n+m$
- e^t : starting time of period $t \in T$
- ℓ^t : ending time of period $t \in T$
- p : unit penalty cost associated with uncollected waste at the end of the planning horizon
- d_j : initial non-negative (deterministic) demand associated with collection point $j = 1, ..., n$
- δ_{jt}^s : non-negative demand associated with collection point $j = 1, ..., n$ which arises at the beginning of period $t \in T$ in scenario $s \in S$.
- M, T_{max}, Q_{max} : large constant

Variables:

- x_{ij}: binary variable equal to 1 if a vehicle travels from node $i \in V'$ to node $j \in V''$ $(j \neq i)$, 0 otherwise (first stage routes)
- U_{jt}: binary variable equal to 1 if node $j \in V''$ is visited in period $t \in T$, 0 otherwise
- $\bar{T}_j$: arrival time at node $j \in V$
- y_{ij}^s: binary variable equal to 1 if a vehicle skips arc going from node $i \in V'$ to node $j \in V''$ $(j \neq i)$ in scenario $s \in S$, 0 otherwise

- Q_j^s: theoretical cumulative load at node $j \in V$ in scenario $s \in S$ (it is obtained by ignoring vehicle capacity)
- R_j^s: quantity of waste exceeding the capacity at node $j = n+1, ..., n+m$ in scenario $s \in S$
- L_j^s: quantity of waste accumulated at collection point $j = 1, ..., n$ in scenario $s \in S$ in periods following its visit, and left over for the next day
- ρ_j^s: binary variable equal to 1 if node $j = 1, ..., n$ is associated with event "capacity saturation" in scenario $s \in S$, 0 otherwise
- α_j^s: binary variable equal to 1 if node $j = 1, ..., n$ is the first node associated with event "capacity saturation" in scenario $s \in S$, 0 otherwise

$$\min \quad \sum_{i \in V'} \sum_{j \in V''(j \neq i)} c_{ij} x_{ij} - \frac{1}{|S|} \sum_{s \in S} \sum_{i=1}^{n} \sum_{j \in V''(j \neq i)} c_{ij} y_{ij}^s + \frac{1}{|S|} \sum_{s \in S} \sum_{j=1}^{n} c_{j0} \alpha_j^s +$$

$$+ \frac{1}{|S|} \sum_{s \in S} \sum_{j=n+1}^{n+m} p R_j^s + \frac{1}{|S|} \sum_{s \in S} \sum_{j=1}^{n} p L_j^s \tag{1}$$

subject to

$$\sum_{j \in V'' \setminus \{i\}} x_{ij} = \sum_{j \in V' \setminus \{i\}} x_{ji} = \sum_{t \in T} U_{it}, \quad i = 1, ..., n \tag{2}$$

$$\sum_{j \in V''} x_{0j} = m \tag{3}$$

$$\sum_{j \in V'} x_{ji} = 1, \quad i = n+1, ..., n+m \tag{4}$$

$$\sum_{t \in T} U_{jt} = 1, \quad j \in V'' \tag{5}$$

$$\bar{T}_0 = 0 \tag{6}$$

$$\bar{T}_j \geq \bar{T}_i + t_{ij} - M(1 - x_{ij}), \quad i \in V', j \in V''(j \neq i) \tag{7}$$

$$e^t - T_{max}(1 - U_{jt}) \leq \bar{T}_j \leq \ell^t + T_{max}(1 - U_{jt}), \quad j \in V'', t \in T \tag{8}$$

$$Q_0^s = 0, \quad s \in S \tag{9}$$

$$Q_j^s \geq Q_i^s - M(1 - x_{ij}) + d_j + \sum_{t \in T} \sum_{\tau=1}^{t} \delta_{j\tau}^s U_{jt}, \quad s \in S, i \in V', j = 1, ..., n (j \neq i) \tag{10}$$

$$Q_j^s \leq Q_i^s + M(1 - x_{ij}) + d_j + \sum_{t \in T} \sum_{\tau=1}^{t} \delta_{j\tau}^s U_{jt}, \quad s \in S, i \in V', j = 1, ..., n (j \neq i) \tag{11}$$

$$Q_j^s \geq Q_i^s - M(1 - x_{ij}), \quad s \in S, i \in V', j = n+1, ..., n+m \tag{12}$$

$$Q_j^s \leq Q_i^s + M(1 - x_{ij}), \quad s \in S, i \in V', j = n+1, ..., n+m \tag{13}$$

$$R_j^s \geq Q_j^s - C, \quad s \in S, j = n+1, ..., n+m \tag{14}$$

$$R_j^s \geq 0, \quad s \in S, j = n+1, ..., n+m \tag{15}$$

$$L_j^s = \sum_{t \in T} \sum_{\tau=t+1}^{|T|} \delta_{j\tau}^s U_{jt}, \quad s \in S, j = 1, ..., n \tag{16}$$

$$\rho_j^s \geq \frac{Q_j^s - C}{Q_{max}}, \quad s \in S, j = 1, \ldots, n \tag{17}$$

$$\rho_j^s \leq 1 - \frac{C - Q_j^s}{Q_{max}}, \quad s \in S, j = 1, \ldots, n \tag{18}$$

$$y_{ij}^s \leq \frac{1}{2}(x_{ij} + \rho_i^s), \quad s \in S, i = 1, ..., n, j \in V''(j \neq i) \tag{19}$$

$$y^s_{0j} = 0, \quad s \in S, \quad j \in V'' \tag{20}$$

$$\alpha^s_j = \sum_{i \in V''} y^s_{ji} - \sum_{i \in V'} y^s_{ij}, \quad s \in S, \quad j = 1, \ldots, n \tag{21}$$

$$x_{ij} \in \{0, 1\}, \quad i \in V', j \in V''(j \neq i) \tag{22}$$

$$U_{jt} \in \{0, 1\}, \quad j \in V'', t \in T \tag{23}$$

$$y^s_{ij} \in \{0, 1\}, \quad s \in S, i \in V', j \in V''(j \neq i) \tag{24}$$

$$\rho^s_j, \alpha^s_j \in \{0, 1\}, \quad s \in S, j = 1, ..., n \tag{25}$$

The objective function is reported in (1). It consists in first stage travel costs minus average cost savings due to partial routes skipping. In addition, it considers penalty cost due to uncollected waste. Equations (2) represent flow conservation constraints. The number of routes starting from depot is bounded in equation (3). It coincides with the number of available vehicles. Equations (4) ensure that routes end into the depot. Note that $x_{0j} = 1$ for some $j = n + 1, ..., n + m$ denotes an empty route. Every node in V'' must be visited in exactly one period as stated by constraints (5). Note that visiting nodes $n + 1, ..., n + m$ within the planning horizon indirectly means fixing a maximum temporal length for the routes. Arrival times at node 0 and at the other nodes are tracked by constraints (6) and (7), respectively. These constraints also eliminate subtours. Constraints (46) allow to identify in which period the nodes in V'' are visited. Every vehicle starts empty from the depot as stated by constraints (47). The theoretical cumulative load at the other nodes in every scenario, obtained by ignoring the vehicle capacity, is defined by means of constraints (48)-(13). In particular, constraints (49)-(13) establish the theoretical cumulative load at the end of every route. Note that the arrival times and the theoretical cumulative loads are non-negative without imposing other constraints. The quantity of waste exceeding the capacity in every route and in every scenario is defined by constraints (50); it cannot be negative according to constraints (15). The demand left over at every collection point after its visit is computed through constraints (16). It is non-negative without imposing other constraints. Constraints (17)-(18) allow

to identify collections points associated with event "capacity saturation". If a vehicle is fully filled at a node, the remaining ones are skipped as implied by constraints (19). Since the "initial" depot cannot be a node associated with event "capacity saturation", constraints (20) avoid that the vehicles skip arcs going from 0. Constraints (21) check whether a collection point is the first node in the route in which we experience a lack of capacity (i.e., it is the first node associated with event "capacity saturation"). Constraints (51)-(41) state the nature of binary variables.

2.2. Oracle Model with Skipping

We need to define the following additional decision variables:

- $\bar{Q}_j$: cumulative load at node j
- D_j: demand collected at node j

Furthermore we define the constant $\bar{D}_j$ which represents the cumulative demand of node j across all the periods, such as $\bar{D}_j = d_j + \sum_{t \in T} \bar{\delta}_j^t$, where $\bar{\delta}_j^t$ represents the actual amount of waste arrived at node j on time-slot t, which in the oracle model is supposed to be known.

$$\min \sum_{i \in V} \sum_{j \in V} c_{ij} x_{ij} + \sum_{i \in V} p(\bar{D}_i - D_i) \tag{26}$$

$$\sum_{j \in V} x_{ij} = \sum_{j \in V} x_{ji} = W_i, \quad i \in V \setminus \{0\} \tag{27}$$

$$\sum_{t \in T} U_{it} \geq D_i / \bar{D}_i, \quad i \in V \setminus \{0\} \tag{28}$$

$$\bar{Q}_j \geq \bar{Q}_i + D_j - C(1 - X_{ij}), \quad i \in V \setminus \{0\} \tag{29}$$

$$0 \leq \bar{Q}_j \leq C, \quad i \in V \setminus \{0\} \tag{30}$$

$$e^t - T_{max}(1 - U_{jt}) \leq \bar{T}_j \leq \ell^t + T_{max}(1 - U_{jt}), \quad j \in V, t \in T \tag{31}$$

$$D_j \leq d_j + \sum_{t \in T} \sum_{\tau = t}^{|T|} \delta_{j\tau}^s U_{jt}, \quad i \in V \setminus \{0\} \tag{32}$$

$$\sum_{t \in T} U_{jt} \leq 1, \quad j \in V, t \in T \tag{33}$$

The objective function, reported in (26) aims at minimizing the fleet routing costs plus the penalties for unserved demand. To collect waste from a node, we have to visit it, as implied by the combination of constraints (27) and (28). The load of the vehicle is tracked by constraints (29) and can never exceed the maximum allowed load, or be negative ((30)). The timeslot in which a node is visited is identified by constraints (31) and We cannot collect more waste than that actually present at a node in the timeslot in which we visit it ((32)). Finally, if we collect waste from a node than we have to visit it in exactly one timeslot, otherwise, we do not visit it. ((33)).

3. Valid inequalities

$$|V'| \leq \sum_{i \in V} \sum_{j \in V} x_{ij} \leq |V'| + m - 1 \tag{34}$$

$$\sum_{i \in V'} \sum_{j \in V''} y_{ijs} \leq \sum_{j \in V''} \rho_{js} \quad s \in S \tag{35}$$

4. Toy instances

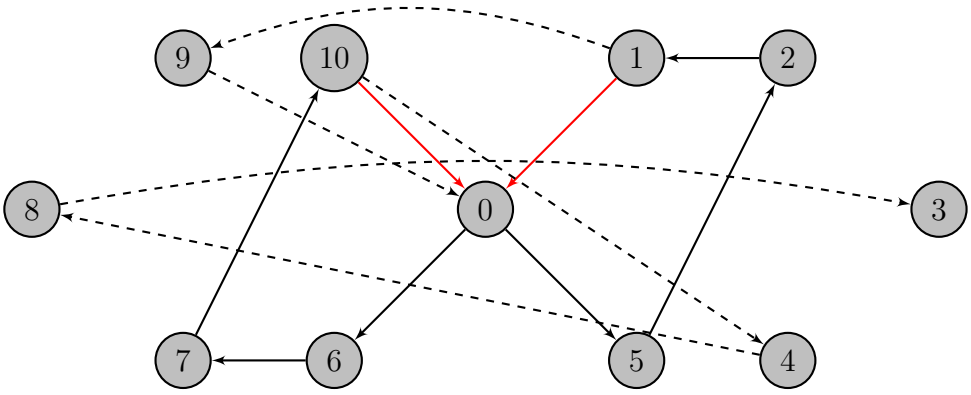


Figure 1: Result for instance I2. Dashed arcs consider $y^s_{i,j} = 1$, while red lines are the arcs connecting the vehicles to the depot

5. Idee per policies

- **P1:** solve a standard VRP with demand equal to the initial demand (completely neglecting future demand)
- **P2:** solve a standard VRP with demand equal to the initial demand + expected value of the sum of the additional demands delta. More conservative policy
- **P3:** solve a standard VRP with demand equal to the initial demand + expected value of the sum of the additional demands up to the central time slot (if we have 3 timeslots we sum up the expected delta of period 1 and 2). This is less conservative than P2, since consider that we are unlikely to visit all the customers in the last timeslot
- **P4:** solve a standard VRP with demand equal to the initial demand plus the expected value of the sum of the delta (like in P2) but with vehicle capacity reduced by x% (this allow to better distribute the load over the vehicles and to have a cushion to be used for unexpected extra demand)
- **P5:** solve a standard VRP with demand equal to the initial demand plus the expected value of the sum of the delta (like in P2) in which half of the vehicle have regular capacity and the other half have reduced capacity by 2x%
- **P6:** solve a VRP with demand equal to the initial demand plus the expected value of the sum of the delta (like in P2) but with a modified objective function in which we minimize the sum of the traveled arc multiplied by the position of the final node in the route (the goal is to insert the farthest nodes at the end of the routes, so in case of skipping we skip the longest arcs).

$$\min \quad \sum_{j \in V''} \frac{1}{d_{0j}} \cdot u_j \tag{36}$$

adding the following constraints to determine the position of a node in the route (u_j).

$$u_j \geq u_i + 1 - (|V'| - 1)(1 - x_{ij}) \quad \forall j \in V' - 0 \quad \forall i \in V' \tag{37}$$

$$u_0 = 0 \tag{38}$$

6. Numerical Results

6.1. Instances

we considered different instances, in order to test the performance both the model and the policies. The stochastic model is compared against the same model where there is only one scenario where the demand accumulation is equal to the average of all scenario, $\bar{\delta}_{jt} = \frac{\sum_s \delta^s_{jt}}{S}$. Then, the solution of this "average" model is evaluated in the original model. We call this solution as the EEV solution (Expected Value).

Different sets of instances has been generated. The first one, denoted with the prefix S, are randomically generated with customers uniformly distributed on a square of 10 units of length. Other instaces are generated using a realistic case where customers are located on map. S instances have the following parameters (Table 1):

Parameter	Value
n	[5,7,10]
m	3
Depot	[Central, Located lower left]
C	[100, 150, 200]
$\|T\|$	3
$\|S\|$	5
$Tmax$	[30, 50, 100]
Average D	50
Average δ	10

Table 1: Basic parameters for random instances named IS2

Starting for the above parameters, demand d_j and accumulation δ^s_{jt} for each customer j has been generated assigning to each customer a type. Different type of customers have different pattern of demand d and accumulation δ over the time slots. Customer types have the following characteristics:

type 1 d low; δ high, low, low

type 2 d average; δ average, high, low

type 3 d high; δ low, low, average.

Table 2: Instance I2S1, Tmax=100, comparison among stochastic solution and EEV solution by varrying capacity of vehicles

Model	Stochastic	EEV	Stochastic	EEV	Stochastic	EEV
C	100	100	150	150	200	200
obj f.	1840.29	1845.04	648.80	657.08	23.71	5.77
Z1	25.46	24.92	23.71	22.78	23.71	21.53
Z2	9.85	8.52	0.48	0.00	0.00	0.00
Z3	3.55	1.35	0.48	0.00	0.00	0.00
Z4	1821.13	1670.21	625.09	199.44	0.00	4.24
Z5	0.00	157.08	0.00	434.86	0.00	0.00

When a demand or an accumulation is high, it is multiplied by a factor of 1.5. Conversely a low demand or accumulation is multiplied by a factor of 0.5. Average demand is left unchanged with respect to the baseline defined in table 1. Then assigned demand or accumulation is the result of a random uniform generation between 0.5 and 1.5 times the resulting demand or accumulation.

6.2. Test of the stochastic model

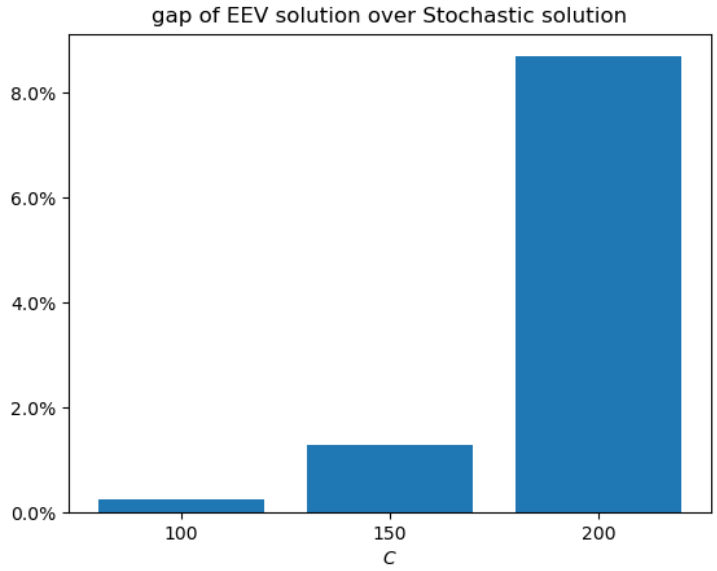


Figure 2: Instance I2S1, Tmax=100, comparison among stochastic solution and EEV solution by varrying capacity of vehicles

7. VRP model for policies

$$\min \quad \sum_{i \in V'} \sum_{j \in V''(j \neq i)} c_{ij} x_{ij} + K \sum_{j} \mu_j \tag{39}$$

subject to

$$\sum_{j \in V'' \setminus \{i\}} x_{ij} = \sum_{j \in V' \setminus \{i\}} x_{ji} = \sum_{t \in T} U_{it}, \quad i = 1, ..., n \tag{40}$$

$$\sum_{j \in V''} x_{0j} = m \tag{41}$$

$$\sum_{j \in V'} x_{ji} = 1, \quad i = n+1, ..., n+m \tag{42}$$

$$\sum_{t \in T} U_{jt} = 1, \quad j \in V'' \tag{43}$$

$$\bar{T}_0 = 0 \tag{44}$$

$$\bar{T}_j \geq \bar{T}_i + t_{ij} - M(1 - x_{ij}), \quad i \in V', j \in V''(j \neq i) \tag{45}$$

$$e^t - T_{max}(1 - U_{jt}) \leq \bar{T}_j \leq \ell^t + T_{max}(1 - U_{jt}), \quad j \in V'', t \in T \tag{46}$$

$$Q_0 = 0 \tag{47}$$

$$Q_j \geq Q_i - M(1 - x_{ij}) + d_j, \quad i \in V', j = 1, ..., n(j \neq i) \tag{48}$$

$$Q_j \geq Q_i - M(1 - x_{ij}), i \in V', j = n+1, ..., n+m \tag{49}$$

$$Q_j \leq C + \mu_j, j = n+1, ..., n+m \tag{50}$$

$$x_{ij} \in \{0, 1\}, \quad i \in V', j \in V''(j \neq i) \tag{51}$$

$$U_{jt} \in \{0,1\}, \quad j \in V'', t \in T \tag{52}$$